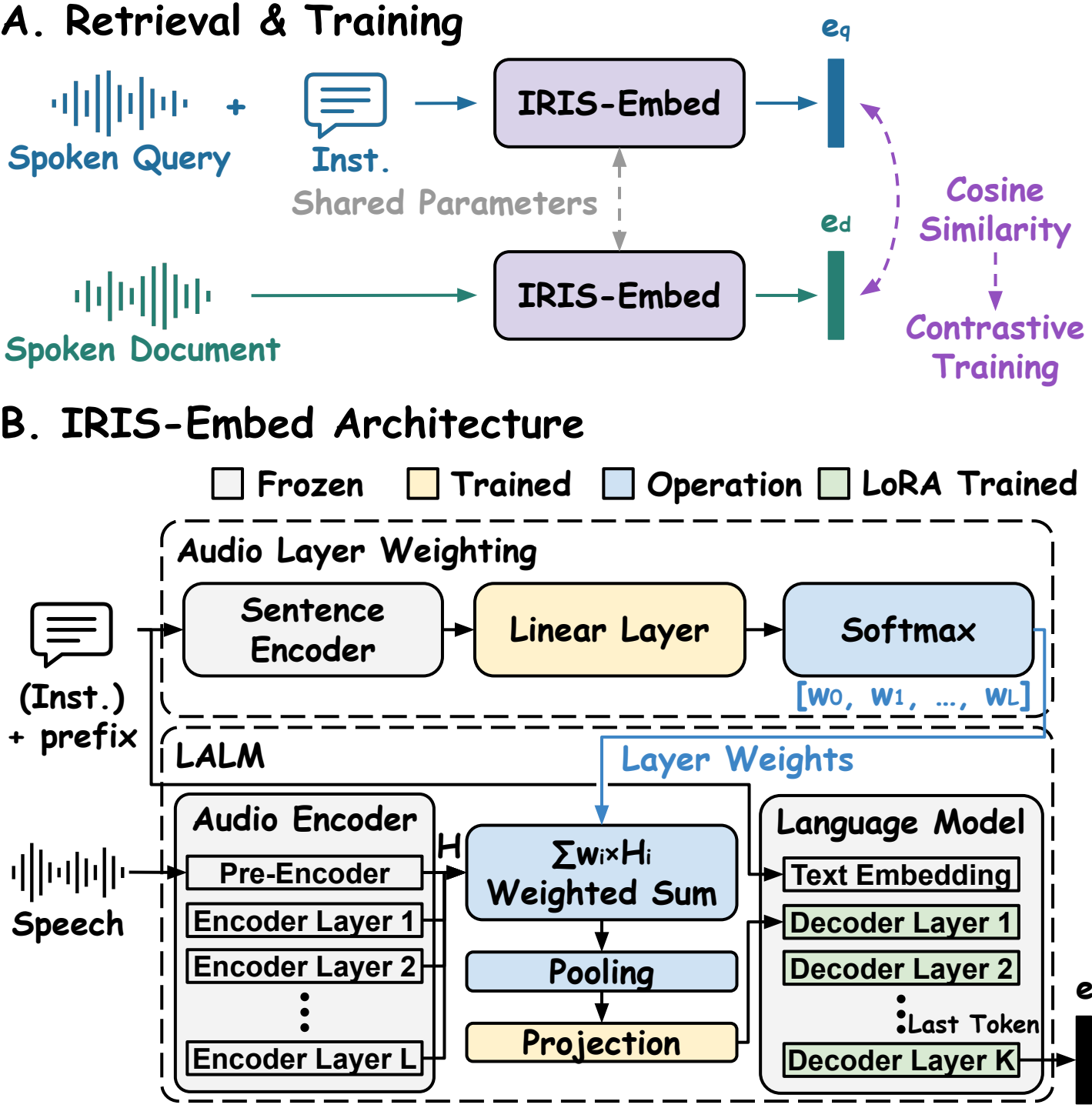

A. Retrieval & Training
Spoken Query
+
Inst.
IRIS-Embed
Shared Parameters
IRIS-Embed
Spoken Document
eq
ed
Cosine Similarity
Contrastive Training
B. IRIS-Embed Architecture
Frozen
Trained
Operation
LoRA Trained
Audio Layer Weighting
Sentence Encoder
Linear Layer
Softmax
[w0, w1, …, wL]
(Inst.) + prefix
LALM
Layer Weights
Audio Encoder
Pre-Encoder
Encoder Layer 1
Encoder Layer 2
Encoder Layer L
H
Speech
∑wi×Hi Weighted Sum
Pooling
Projection
Language Model
Text Embedding
Decoder Layer 1
Decoder Layer 2
Last Token
Decoder Layer K
e